

Complete AI/ML Engineering Roadmap

Phase 1 — Python Programming for AI/ML

Python Fundamentals

Python syntax, Variables and data types, Type conversion, Operators, Input/output, Conditional statements, Loops, break/continue/pass, Functions, Parameters and arguments, *args, **kwargs, Lambda functions, Scope, Recursion

Python Data Structures

List, Tuple, Set, Dictionary, String, List/Dictionary/Set comprehension, Nested data structures, Stack and Queue

Python OOP

Classes, Objects, Constructors, Instance/Class/Static methods, Encapsulation, Inheritance, Polymorphism, Abstraction, Magic methods, Dataclasses

Advanced Python

Exception handling, File handling, Modules, Packages, Virtual environments, pip, Iterators, Generators, Decorators, Context managers, Type hints, asyncio, Logging, Unit testing, pytest

Python for Developers

JSON, CSV, REST API consumption, HTTP requests, Environment variables, .env, Configuration management, Git/GitHub, Package management

Phase 2 — Mathematics for AI/ML

Statistics

Mean, Median, Mode, Variance, Standard deviation, Percentiles, Probability, Conditional probability, Bayes theorem, Normal distribution, Correlation, Covariance, Sampling, Hypothesis testing, Confidence intervals

Linear Algebra

Scalars, Vectors, Matrices, Matrix operations, Dot product, Matrix multiplication, Transpose, Identity matrix, Inverse, Eigenvalues, Eigenvectors

Calculus

Functions, Limits, Derivatives, Partial derivatives, Gradients, Chain rule, Gradient descent

Optimization

Cost function, Loss function, Convex optimization, Gradient descent, Learning rate, Local/global minima, SGD, Mini-batch gradient descent

Phase 3 — Python Data Science Stack

NumPy

Arrays, Dimensions, Shape, Indexing, Slicing, Broadcasting, Vectorization, Matrix operations, Random numbers, Statistical functions, Linear algebra

Pandas

Series, DataFrame, CSV/Excel/JSON, Filtering, Sorting, Missing values, Duplicates, GroupBy, Merge, Join, Concatenation, Pivot tables, Date/Time

Data Visualization

Matplotlib, Seaborn, Plotly, Line, Bar, Histogram, Scatter, Box plot, Heatmap, Correlation matrix

Phase 4 — Data Preparation

Data Cleaning

Missing values, Duplicates, Outliers, Incorrect types, Inconsistent values, Validation

Feature Engineering

Encoding, One-hot, Label encoding, Scaling, Normalization, Standardization, Feature transformation, Feature selection, Dimensionality reduction

Data Splitting

Training, Validation, Test sets, Cross-validation, Stratified splitting, Data leakage

Phase 5 — Machine Learning Fundamentals

ML Fundamentals

AI vs ML, ML vs DL, Training, Inference, Dataset, Features, Labels, Model, Parameters, Hyperparameters, Loss, Optimization, Overfitting, Underfitting, Bias, Variance

Supervised Learning

Linear, Multiple, Polynomial, Ridge, Lasso, Elastic Net, Logistic Regression, KNN, Naive Bayes, Decision Tree, Random Forest, SVM

Unsupervised Learning

K-Means, Hierarchical Clustering, DBSCAN, PCA, Association Rules

Ensemble Learning

Bagging, Boosting, Random Forest, AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost

Phase 6 — ML Evaluation

Regression Metrics

MAE, MSE, RMSE, R^2 , Adjusted R^2 , MAPE

Classification Metrics

Accuracy, Precision, Recall, F1, Confusion Matrix, ROC, AUC, PR-AUC

Model Validation

Train/Test Split, K-Fold, Stratified K-Fold, Grid Search, Random Search, Bayesian Optimization

Phase 7 — Scikit-Learn

Scikit-Learn Mastery

Estimators, Transformers, Pipelines, ColumnTransformer, Preprocessing, Feature Selection, Model Selection, Cross-validation, Hyperparameter tuning, Model persistence

Phase 8 — Deep Learning

Neural Network Fundamentals

Neuron, Perceptron, Weights, Bias, Activation, Forward/Backpropagation, Loss, Gradient descent, Epoch, Batch

Architectures

ANN, CNN, RNN, LSTM, GRU, Autoencoders, Transformers

Frameworks

PyTorch, TensorFlow/Keras, Hugging Face

Phase 9 — Computer Vision

Image Processing

Pixels, RGB, Grayscale, Resizing, Normalization, Augmentation

Computer Vision

Classification, Object Detection, Segmentation, Transfer Learning, YOLO, OpenCV

Phase 10 — NLP

NLP Fundamentals

Tokenization, Stop words, Stemming, Lemmatization, N-grams, Bag of Words, TF-IDF

NLP ML

Text Classification, Sentiment Analysis, Spam Detection, NER, Text Similarity

Modern NLP

Word2Vec, GloVe, Transformers, Attention, BERT

Phase 11 — Generative AI

Generative AI

Foundation Models, LLMs, Tokens, Embeddings, Context Window, Parameters, Temperature, Top-K, Top-P

LLM Architecture

Transformer, Self-Attention, Multi-head Attention, Positional Encoding, Encoder, Decoder

LLM APIs

Authentication, Prompt/Response, Streaming, Structured Output, Tool Calling, JSON, Token Usage, Rate Limits

Phase 12 — Prompt Engineering

Prompt Engineering

Zero-shot, Few-shot, Role prompting, System/User messages, Structured prompting, Output constraints, Prompt templates, Evaluation

Phase 13 — Embeddings & Vector DB

Embeddings

Semantic similarity, Cosine similarity, Chunking

Vector Databases

FAISS, Chroma, Qdrant, Pinecone, Weaviate, Milvus

Phase 14 — RAG

RAG

Ingestion, Chunking, Embeddings, Retrieval, Context, Generation, Metadata filtering, Reranking

Advanced RAG

Hybrid retrieval, Multi-query, Parent-child, Hallucination reduction

Phase 15 — AI Agents

Agent Fundamentals

Tools, Memory, Planning, Execution, Observation

Architectures

Single, Multi-agent, Workflow, Human-in-the-loop

Frameworks

LangChain, LangGraph, LlamaIndex

Phase 16 — AI Engineering with Python

FastAPI

REST APIs, Pydantic, JWT, Async APIs, File uploads, Background tasks

Phase 17 — AI Databases

SQL

PostgreSQL, MySQL, CRUD, Joins, Indexing, Transactions

NoSQL

MongoDB, Redis

Vector DB

Qdrant, Pinecone, FAISS

Phase 18 — MLOps

Model Lifecycle

Training, Evaluation, Registry, Deployment, Monitoring

MLflow

Experiment tracking, Metrics, Artifacts

Versioning

Git, DVC

Phase 19 — Docker & Deployment

Docker

Images, Containers, Dockerfile, Compose, Volumes, Networks

CI/CD

GitHub Actions, Testing, Build Pipelines

Phase 20 — AWS for AI/ML

AWS Fundamentals

IAM, EC2, S3, RDS, VPC, CloudWatch

AWS ML

SageMaker, Endpoints, Monitoring

AWS GenAI

Foundation Models, RAG, AI Agents

Phase 21 — Production AI

AI System Design

Scalability, Caching, Load Balancing, Queues, Rate Limiting, Authentication, Monitoring

Phase 22 — LLMOps

LLM Production

Prompt versioning, Evaluation, Cost, Latency, Guardrails, Observability

Phase 23 — AI Security

AI Security

Prompt injection, Jailbreak awareness, Data leakage, API key security, Authentication, Authorization, Rate limiting, Input/Output validation